

Exp-5- Record-Implementation of Filters

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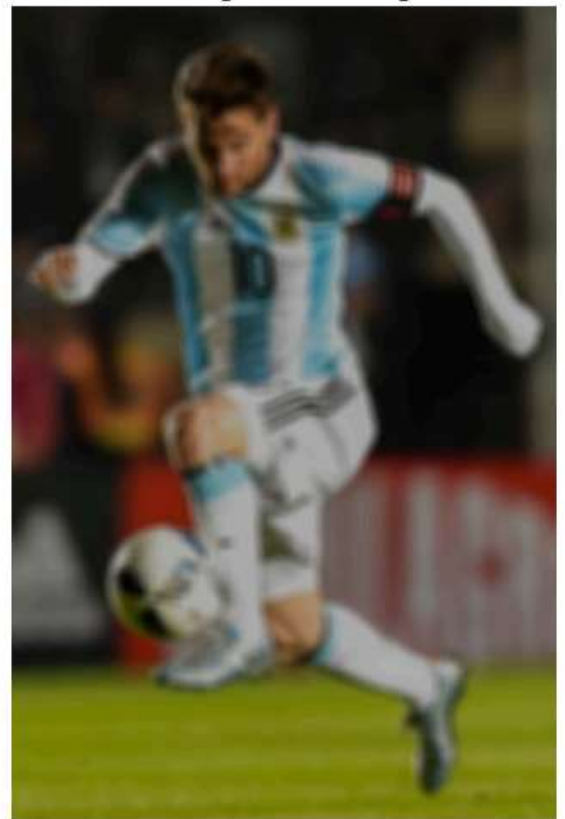
In [1]:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np
image1=cv2.imread("Football.jpg")
image2=cv2.cvtColor(image1,cv2.COLOR_BGR2RGB)
kernel=np.ones((11,11),np.float32)/169
image3=cv2.filter2D(image2,-1,kernel)
plt.figure(figsize=(9,9))
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(image3)
plt.title("Average Filter Image")
plt.axis("off")
plt.show()
```

Original Image



Average Filter Image



```
In [2]: kernel1=np.array([[1,2,1],[2,4,2],[1,2,1]])/16
image2=cv2.cvtColor(image1,cv2.COLOR_BGR2RGB)
image3=cv2.filter2D(image2,-1,kernel1)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(image3)
plt.title("Weighted Average Filter Image")
plt.axis("off")
plt.show()
```

Original Image



Weighted Average Filter Image



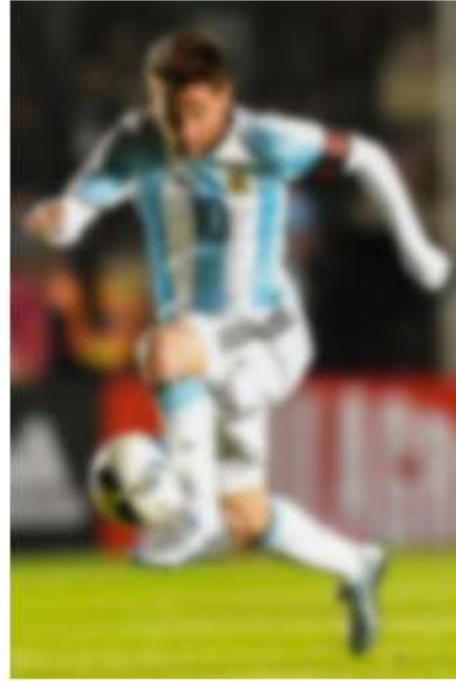
In [3]:

```
gaussian_blur=cv2.GaussianBlur(image2,(33,33),0,0)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(gaussian_blur)
plt.title("Gaussian Blur")
plt.axis("off")
plt.show()
```

Original Image



Gaussian Blur



In [4]:

```
median=cv2.medianBlur(image2,13)
plt.figure(figsize=(9,9))
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(median)
plt.title("Median Blur")
plt.axis("off")
plt.show()
```

Original Image



Median Blur



```
In [5]: kernel2=np.array([[ -1, -1, -1],[2, -2, 1],[2, 1, -1]])  
image3=cv2.filter2D(image2, -1,kernel2)  
plt.subplot(1,2,1)  
plt.imshow(image2)  
plt.title("Original Image")  
plt.axis("off")  
plt.subplot(1,2,2)  
plt.imshow(image3)  
plt.title("Laplacian Kernel")  
plt.axis("off")  
plt.show()
```

Original Image



Laplacian Kernel




```
In [6]: laplacian = cv2.Laplacian(image2, cv2.CV_64F)
plt.subplot(1, 2, 1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1, 2, 2)
laplacian_abs = np.uint8(np.absolute(laplacian))
plt.imshow(laplacian_abs, cmap='gray')
plt.title("Laplacian Operator")
plt.axis("off")
plt.show()
```

Original Image



Laplacian Operator

